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# LLC Converter Design for a Photovoltaic Micro-Inverter

*Internship Report*

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# 1 Abstract

This report investigates the design of a GaN-based LLC step-up converter as the first power stage of a photovoltaic micro-inverter. The primary objective is to optimize both efficiency and performance, with particular emphasis on maximizing energy conversion and minimizing power losses. In the current landscape of renewable energy, the micro-inverter plays a critical role in transforming solar energy into usable AC electricity. The adoption of Gallium Nitride (GaN) technology proves highly promising due to its high-speed switching capability and low on-state resistance, offering significant performance improvements over conventional silicon-based solutions.

The methodology includes a detailed assessment of circuit requirements — input voltage, output voltage, switching frequency, and maximum current — followed by an evaluation of commercially available GaN devices. The design phase involves determining the key parameters of the LLC step-up circuit, including resonant inductance, resonant capacitance, and magnetizing inductance. Special attention is devoted to maximizing efficiency and minimizing both switching and conduction losses.

Circuit performance is validated through detailed simulations in LTspice and compared against design specifications to verify compliance with efficiency and performance targets. The findings are discussed in terms of their practical implications for the deployment of GaN technology in power conversion systems, highlighting its potential to improve the overall performance of micro-inverters and accelerate the adoption of solar energy. This report provides a comprehensive reference for future designs in the field of renewable energy conversion.

# 2 Introduction

## 2.1 Photovoltaic Systems

Photovoltaic (PV) systems represent a fundamental component of the modern energy landscape, offering a sustainable and scalable solution for electrical power generation. These systems exploit the photovoltaic effect — the direct conversion of sunlight into electrical energy through semiconductor devices known as solar cells. Their relevance has grown significantly in response to the increasing global demand for clean and renewable energy, combined with the urgent need to reduce greenhouse gas emissions and mitigate climate change.

PV systems consist of several key elements. At the core are photovoltaic cells, typically manufactured from crystalline or amorphous silicon, which generate direct current (DC) when illuminated. Individual cells are connected in series or parallel to form photovoltaic modules, or solar panels. Multiple modules can be combined into arrays to scale up the generated power. Additional essential components include power inverters, which convert the DC produced by the cells into alternating current (AC) for grid injection or local consumption. Systems may also incorporate battery storage, monitoring and control devices, and structural mounting hardware.

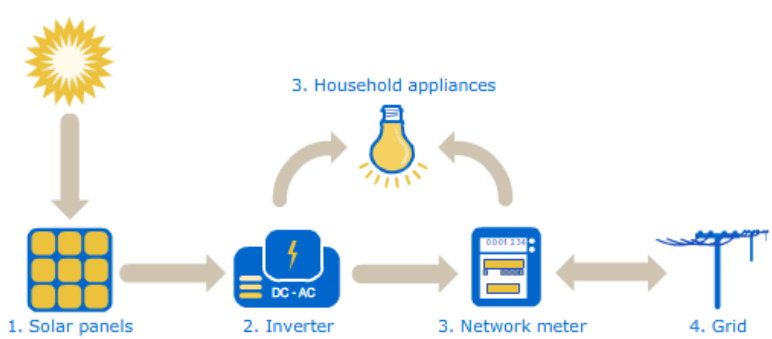



Figure 1: Overview of a photovoltaic system.

System operation begins with solar irradiance striking the PV cells and setting electrons in motion, thereby generating a DC current. This current is fed to inverters for AC conversion before being consumed locally or exported to the grid. During periods of low or no irradiance — such as nighttime — battery storage systems can sustain a continuous power supply.

Key performance metrics include the peak power rating, expressed in Watt-peak ($W_p$), representing maximum output under standard test conditions (STC), and the overall conversion efficiency, which quantifies the fraction of incident solar energy successfully converted to electricity. Panel orientation, tilt angle, and surface cleanliness all influence system efficiency. Despite a broad range of applications — residential, commercial, industrial, and agricultural — widespread adoption still faces challenges including high upfront costs, weather dependence, the need for energy storage to buffer generation variability, and regulatory constraints.

### 2.2 Inverter Topologies

Inverters are essential for converting the DC power generated by PV panels into AC power suitable for grid connection or local loads. Several inverter architectures are used in PV installations, each with specific characteristics and trade-offs.

**Central inverters** are among the most established solutions for large-scale PV plants. A single high-power inverter is connected to the entire PV array, receiving energy from all panels wired in series–parallel combinations and converting it from DC to high-voltage AC. While cost-effective at scale, they suffer from reduced efficiency under partial shading or panel mismatch.

**String inverters** represent an intermediate solution. Multiple inverters are distributed throughout the installation, each handling the output of a series string of panels. This reduces the impact of partial shading on the entire array but still limits individual panel optimization.

**Micro-inverters** are the most advanced and granular topology. A dedicated inverter is assigned to each individual panel (or pair of panels), performing DC-to-AC conversion at the module level. This architecture enables independent maximum power point tracking (MPPT) for each panel, maximizing energy harvest under partial shading or non-uniform conditions. Additional benefits include enhanced fault isolation, improved monitoring granularity, and increased system reliability. The primary trade-offs are higher per-unit costs and more complex installation and maintenance logistics compared to string and central solutions.

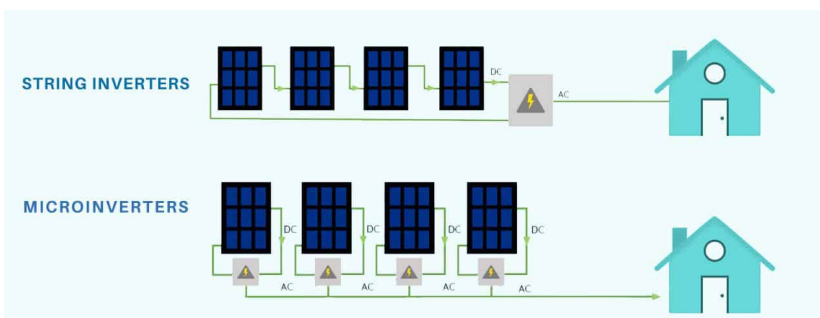



Figure 2: String inverter vs. micro-inverter architecture.

### 2.3 GaN Transistors

Ongoing research in power semiconductor devices has driven the adoption of Gallium Nitride (GaN) as an alternative to Silicon (Si) for power MOSFET fabrication. GaN exhibits several material properties that make it highly attractive for high-performance, high-efficiency power conversion: a wider bandgap, higher electron mobility, and higher critical electric field strength.

The operating principle of a GaN MOSFET is analogous to that of a Si MOSFET. The conduction

channel is formed in the GaN substrate, while the gate dielectric is typically a high-$\kappa$ material such as aluminium oxide ($Al_2O_3$). Gate voltage modulates the carrier density in the channel, controlling current flow. Crucially, GaN's wide bandgap allows it to withstand high electric fields without avalanche breakdown, while its superior carrier mobility enables low on-state resistance ($R_{\mathrm{DS(on)}}$) at reduced die areas.

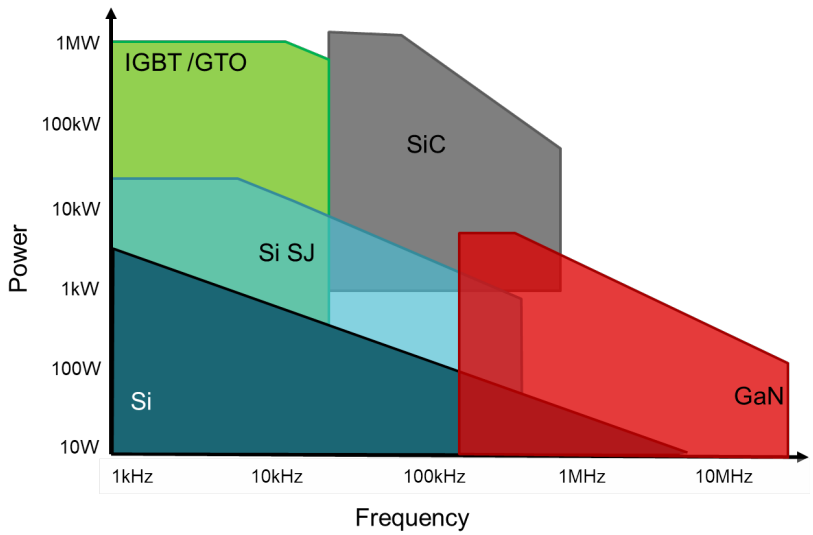



Figure 3: Material property comparison: Si, SiC, and GaN.

Key advantages of GaN MOSFETs over Si devices include:

- **High switching speed:** Superior carrier mobility and lower gate capacitance allow significantly faster switching transitions, reducing switching losses and enabling operation at higher frequencies.
- **Low on-state resistance:** Intrinsically lower resistivity translates directly into reduced conduction losses and improved efficiency.
- **High breakdown voltage:** GaN supports higher operating voltages without entering breakdown, making it suitable for high-power applications.
- **High operating temperature:** Improved thermal stability allows GaN devices to operate reliably at elevated junction temperatures.
- **Compact form factor:** GaN dies can be made significantly smaller than Si equivalents for the same voltage and current ratings, enabling denser power converters.

GaN MOSFETs are now widely adopted in DC-DC converters, DC-AC inverters, high-frequency switching power supplies, wireless charging systems, automotive powertrains, aerospace, and medical electronics.

## 3 DC-DC Step-Up Converter Topologies

### 3.1 Topology Overview

DC-DC converters can be broadly classified by the presence or absence of galvanic isolation between input and output. **Non-isolated** converters share a common ground between input and output, while **isolated** converters employ a transformer or coupled inductor to provide electrical separation. The choice between isolated and non-isolated topologies depends primarily on safety requirements and the need for galvanic separation.

Among non-isolated topologies, the **boost converter** is the most common step-up solution. Simple to design and implement, it suffers from efficiency limitations, particularly at high switching frequencies, where semiconductor switching losses become dominant.

Among isolated topologies, the **flyback converter** uses a coupled inductor/transformer and is well-suited for low-to-medium power applications requiring galvanic isolation. The **LLC resonant converter** is a more sophisticated solution that exploits resonant switching to achieve *soft switching* — significantly improving efficiency by eliminating hard-switching losses.

The fundamental distinction between hard switching and soft switching lies in the switching conditions:

- In **hard switching**, devices switch while carrying significant current, generating high peak power dissipation and electromagnetic interference (EMI).
- In **soft switching**, devices switch at or near zero voltage (ZVS) or zero current (ZCS), eliminating capacitive discharge losses and dramatically reducing EMI.

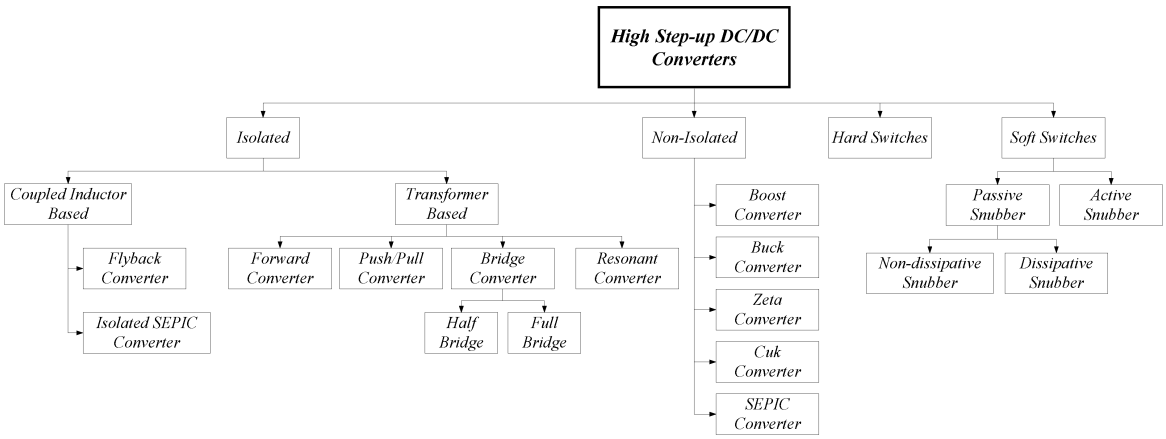



Figure 4: DC-DC converter topology classification.

### 3.2 Efficiency Comparison

Converter efficiency is defined as the ratio of output power to input power and is a critical figure of merit for energy conversion systems. Higher efficiency implies lower thermal dissipation and better overall system sustainability. Efficiency depends on load, input voltage, switching frequency, and component quality.

Table 1 summarizes the efficiency of the most relevant isolated high step-up DC-DC topologies, based on the study *"Topologies and Design Characteristics of Isolated High Step-Up DC–DC Converters for Photovoltaic Systems"* [1].

Table 1: Efficiency comparison of isolated high step-up DC-DC converters.

| **Converter Topology** | **Eff. (%)** | $V_{in}$ **(V)** | $V_{out}$ **(V)** | $f_{sw}$ **(kHz)** | $P_{out}$ **(W)** |
|---|---|---|---|---|---|
| High Step-Up Forward | 86.9 | 12 | 100 | 50 | 100 |
| High Step-Up Push-Pull | 92.7 | 22 | 200 | 100 | 250 |
| High Step-Up Isolated SEPIC | 93.8 | 48 | 200 | 50 | 100 |
| High Step-Up Dual Flyback | 95.3 | 24 | 200 | 50 | 120 |
| **High Step-Up LLC** | **97.5** | **25** | **500** | **113** | **250** |

The LLC converter achieves the highest efficiency among all evaluated topologies, making it the preferred choice for this application.

# 4 LLC Converter

## 4.1 Operating Principle

The LLC converter is a DC-DC topology that leverages high-frequency resonance to achieve optimized efficiency, reduced component sizes, and lower cost. In this step-up configuration, it boosts the PV panel voltage ( 24 V) to a 400 V DC bus suitable for the downstream inverter stage.

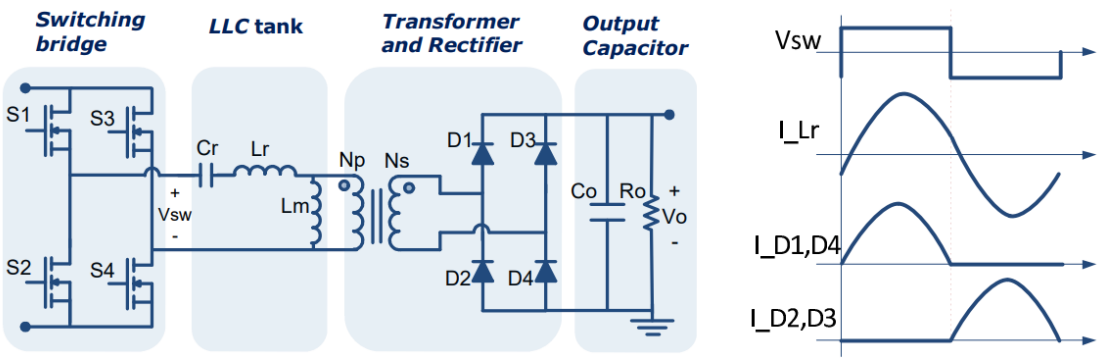
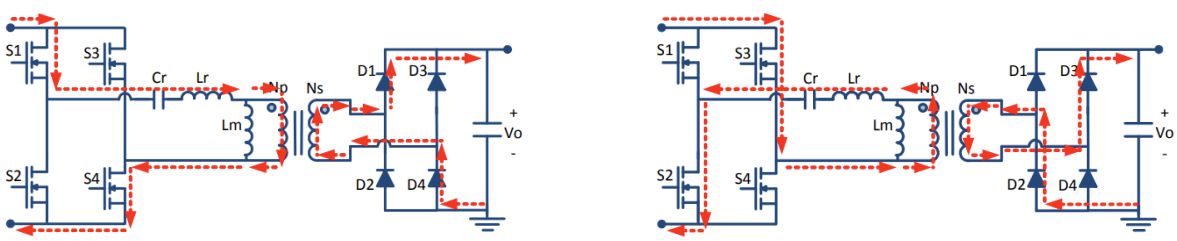



Figure 5: Full-bridge LLC converter schematic.

The switching bridge generates a square-wave voltage that excites the LLC resonant tank (composed of $L_r$, $L_m$, and $C_r$). The resonant tank produces a near-sinusoidal current that is scaled by the transformer and rectified by the secondary-side bridge. An output capacitor filters the rectified current to produce a stable DC output voltage.

The converter exhibits two key operating sub-intervals per switching half-cycle:

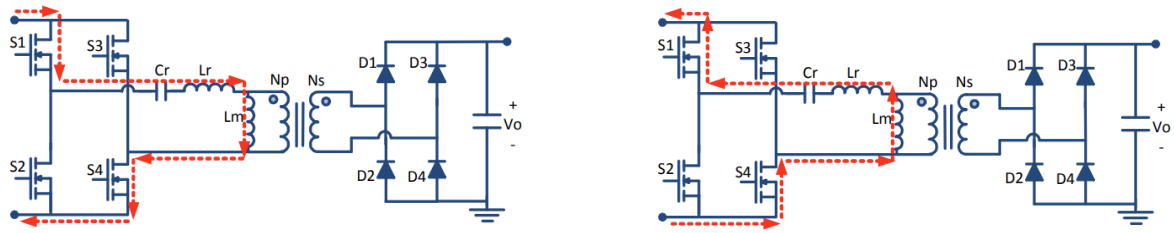
- **Power transfer interval:** The resonant tank is excited by a positive (or negative) voltage. Resonant current flows through the transformer, transferring energy to the secondary side. The magnetizing inductor voltage clamps to the reflected output voltage, and the magnetizing current charges or discharges linearly. The net current delivered to the secondary is the difference between the resonant current and the magnetizing current.




(a) Positive half-cycle power transfer. (b) Negative half-cycle power transfer.

Figure 6: Power transfer equivalent circuits.

- **Freewheeling interval:** This occurs when $f_s < f_r$. Once the resonant current decays to the magnetizing current level, the secondary diodes stop conducting and the magnetizing inductor enters resonance with $L_r$ and $C_r$ at a lower sub-resonant frequency. During this interval, primary current is approximately constant and circulates in the primary loop without transferring energy to the load.




(a) Positive half-cycle freewheeling. (b) Negative half-cycle freewheeling.

Figure 7: Freewheeling equivalent circuits.

The LLC converter can be implemented as a half-bridge (voltage gain = 0.5) or full-bridge (voltage gain = 1.0). The full-bridge configuration is adopted here for its superior efficiency.

The voltage gain of the resonant tank is expressed as:

$$K(Q,\, m,\, F_x) = \left| \frac{V_{o,\mathrm{ac}}(s)}{V_{in,\mathrm{ac}}(s)} \right| = \frac{F_x^2(m-1)}{\sqrt{\left(mF_x^2-1\right)^2 + F_x^2\left(F_x^2-1\right)^2(m-1)^2Q^2}}$$

where $F_x = f_s/f_r$ is the normalized switching frequency, $Q$ is the quality factor, and $m = (L_r + L_m)/L_r$ is the inductance ratio. This gain function is plotted in MATLAB for multiple values of $Q$ and $m$:

```
clc; clear all; close all;

m = [3 6 15];
wx = logspace(-1, 1, 1000);
colors = ['r' 'g' 'b' 'c' 'm' 'k'];
Q = [0.2 0.3 0.5 0.7 1 5];

for y = 1:3
    figure
    for i = 1:6
        Gain = (((8/(pi^2)) * (wx.^2) * (m(y)-1)) ./ ...
            (((m(y)*(wx.^2)-1).^2 + ...
             ((wx.^2).*((wx.^2)-1).^2).*((m(y)-1).^2)*(Q(i)^2)).^0.5));
        semilogx(wx, Gain, 'Color', colors(i), 'LineWidth', 3)
        hold on
    end
    legend('Q=0.2','Q=0.3','Q=0.5','Q=0.7','Q=1','Q=5')
    title(['|G|(\omega) vs \omega  |  m = ', num2str(m(y))], 'FontSize', 18)
    xlabel('\omega_x', 'FontSize', 18)
    ylabel('|G| (\omega_x)', 'FontSize', 18)
    grid on
end
```

For this design, $m = 6$ is selected as the baseline, subject to further optimization during the sizing process. As shown in Figure 8, lower-$Q$ curves correspond to lighter loads while higher-$Q$ curves represent heavier loads. All curves converge at $F_x = 1$ (i.e., $f_s = f_r$), where the gain equals unity. The peaks of the gain curves define the boundary between inductive and capacitive impedance regions of the tank.

**Operating in the inductive region is mandatory throughout the full input voltage and load range**, as Zero Voltage Switching (ZVS) is only achievable there. In the capacitive region, current

leads voltage, causing hard switching of the body diodes, high current spikes, increased losses, and potential device failure.

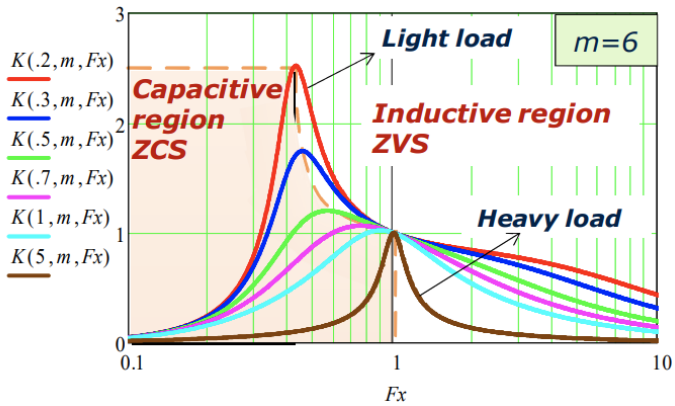
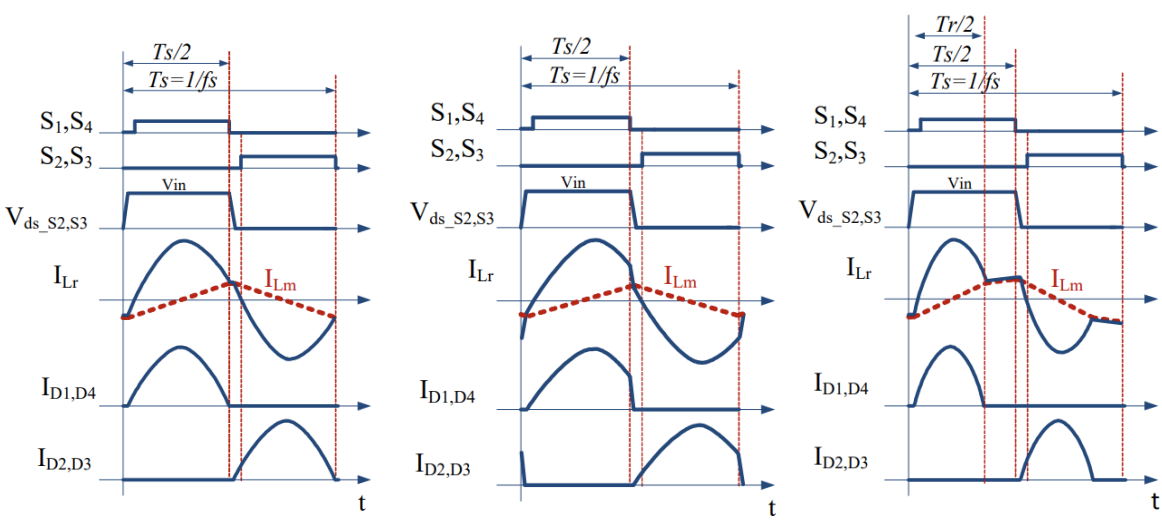



Figure 8: Resonant tank gain vs. normalized frequency (MATLAB).

The converter operates in three distinct modes depending on switching frequency relative to resonant frequency:

1. **At resonance ($f_s = f_r$):** Each switching half-period contains exactly one resonant half-period. Resonant current reaches the magnetizing current at the end of each half-cycle, the secondary current falls to zero naturally (ZCS for diodes), and the transformer turns ratio is set so that the converter delivers nominal gain of unity at this operating point — the highest efficiency condition.
2. **Above resonance ($f_s > f_r$):** The resonant half-period is interrupted before completion. Primary MOSFETs experience increased turn-off losses, while secondary rectifiers undergo hard switching. This mode is used for buck (step-down) regulation when $V_{in}$ exceeds the nominal value.
3. **Below resonance ($f_s < f_r$):** The resonant half-period completes before the switching half-period ends, introducing a freewheeling interval. Increased primary-side circulating current raises conduction losses. This mode is used for boost (step-up) regulation when $V_{in}$ falls below nominal.




(a) $f_s = f_r$ (b) $f_s > f_r$ (c) $f_s < f_r$

Figure 9: Operating mode waveforms.

### 4.2 Design Specifications

PV panels generate a voltage typically ranging from 20 V to 40 V depending on irradiance, temperature, and panel orientation. The nominal input voltage is set to 24 V, a common operating point for residential PV systems. The target output voltage is 400 V DC, providing adequate headroom for the downstream H-bridge inverter to generate a 230 $V_{RMS}$ sinusoidal output while minimizing input current and conduction losses. The rated output power of 400 W reflects a standard commercial micro-inverter power class for residential applications. The switching frequency range of 100 kHz to 1 MHz is explored to characterize the efficiency–size trade-off.

Table 2: Design specifications.

| **Parameter** | **Value** |
|---|---|
| Output voltage $V_{out}$ | 400 V |
| Input voltage $V_{in}$ | 20 V to 40 V (24 V nominal) |
| Maximum output power $P_{out}$ | 400 W |
| Resonant frequency $f_r$ | 100 kHz–1 MHz |

### 4.3 Component Sizing

The LLC design methodology follows the guidelines provided by Infineon in *"Resonant LLC Converter: Operation and Design"* [2], which prescribes a structured design flow starting from the selection of $Q$ and $m$.

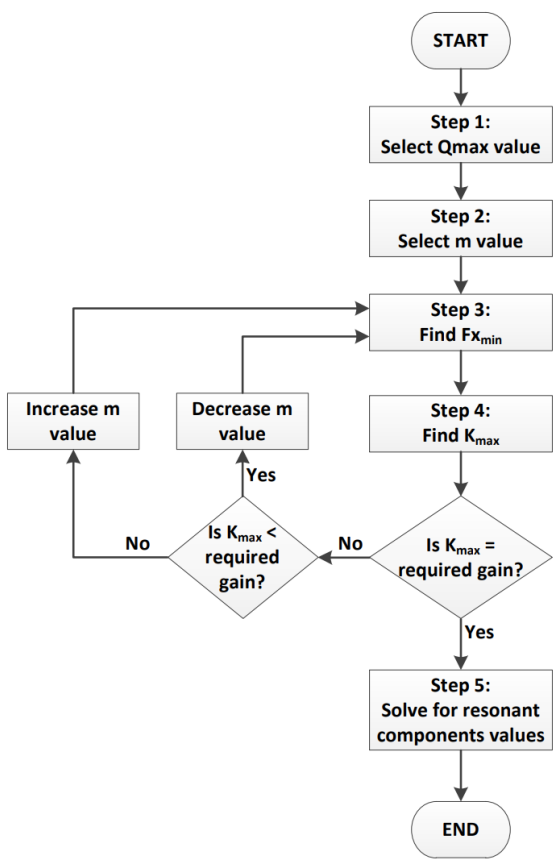



Figure 10: Design flow recommended by Infineon.

The design parameters are selected as follows:

- $Q$: higher values suit higher output current applications; lower values provide better regulation range.
- $m$: lower values increase boost factor but narrow the frequency range; higher values increase $L_m$, reducing magnetizing current and improving efficiency.

After verifying that the maximum gain $K_{max}$ satisfies design requirements (via the MATLAB gain plots), the component values are computed using the following equations:

$$\frac{N_p}{N_s} = \frac{V_{in,nom}}{V_o} \cdot M_{nom} \qquad M_{max} = \frac{V_{in,nom}}{V_{in,min}} \cdot M_{nom} \qquad M_{min} = \frac{V_{in,nom}}{V_{in,max}} \cdot M_{nom}$$

$$R_{ac,min} = \frac{8}{\pi^2} \cdot \frac{N_p^2}{N_s^2} \cdot \frac{V_o^2}{P_{o,max}} \qquad Q_{max} = \frac{\sqrt{L_r/C_r}}{R_{ac,min}}$$

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}} \qquad m = \frac{L_r + L_m}{L_r} \qquad f_{s,min} = F_{x,min} \cdot f_r$$

$F_{x,min}$ and $K_{max}$ are extracted graphically from the MATLAB gain plots. To automate and streamline the sizing process, a dedicated C program was developed:

```c
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

int main() {
    double Vin_min, Vin_max, Vin_nom, Vout, Pout;
    double fr, M_nom, Qmax, m, Fxmin, Kmax;

    printf("Enter Vin_min (V): ");  scanf("%lf", &Vin_min);
    printf("Enter Vin_max (V): ");  scanf("%lf", &Vin_max);
    printf("Enter Vin_nom (V): ");  scanf("%lf", &Vin_nom);
    printf("Enter Vout (V): ");     scanf("%lf", &Vout);
    printf("Enter Pout (W): ");     scanf("%lf", &Pout);
    printf("Enter fr (Hz): ");      scanf("%lf", &fr);
    printf("Enter M_nom (1=Full Bridge, 0.5=Half Bridge): ");
                                    scanf("%lf", &M_nom);
    printf("Enter Qmax: ");         scanf("%lf", &Qmax);
    printf("Enter m: ");            scanf("%lf", &m);
    printf("Enter Fxmin: ");        scanf("%lf", &Fxmin);
    printf("Enter Kmax: ");         scanf("%lf", &Kmax);

    double pi = M_PI;
    double Np_Ns   = (Vin_nom / Vout) * M_nom;
    double M_max   = (Vin_nom / Vin_min) * M_nom;
    double M_min   = (Vin_nom / Vin_max) * M_nom;
    double fs_min  = Fxmin * fr;
    double Qmax_vmin = Qmax * (Vin_min / Vin_max);

    double Rac_min = (8.0 / pow(pi, 2)) * pow(Np_Ns, 2) * (pow(Vout, 2) / Pout);
    double Cr  = sqrt(1.0 / (pow(2*pi*fr, 2) * pow(Qmax*Rac_min, 2)));
    double Lr  = pow(Qmax*Rac_min, 2) * Cr;
    double Lm  = (m * Lr) - Lr;

    printf("\n--- Results ---\n");
```

```
    printf("Np/Ns           : %lf\n",    Np_Ns);
    printf("M_max           : %lf\n",    M_max);
    printf("M_min           : %lf\n",    M_min);
    printf("fs_min (Hz)     : %lf\n",    fs_min);
    printf("Qmax@Vmin       : %lf\n",    Qmax_vmin);
    printf("Kmax            : %lf  |  M_max: %lf\n", Kmax, M_max);
    printf("Rac_min (Ohm)   : %lf\n",    Rac_min);
    printf("Lr (H)          : %.15lf\n", Lr);
    printf("Cr (F)          : %.15lf\n", Cr);
    printf("Lm (H)          : %.15lf\n", Lm);

    printf("\n--- Scaled values ---\n");
    printf("Lr : %.4f nH\n", Lr * 1e9);
    printf("Cr : %.4f nF\n", Cr * 1e9);
    printf("Lm : %.4f nH\n", Lm * 1e9);

    return 0;
}
```

Running the program with the project specifications yields the following optimal component values:

Table 3: Optimal LLC resonant component values.

| $f_{sw}$ | $Q$ | $m$ | $N_p/N_s$ | $f_{s,min}$ | $L_r$ | $C_r$ | $L_m$ |
|---|---|---|---|---|---|---|---|
| 100 kHz | 0.4 | 6.3 | 0.06 | 48.9 kHz | 743 nH | 3.4 µF | 3.93 µH |
| 500 kHz | 0.3 | 6.3 | 0.06 | 218.5 kHz | 111.46 nH | 909 nF | 590.7 nH |
| 1 MHz | 0.2 | 6.3 | 0.06 | 417 kHz | 37.15 nH | 681.77 nF | 196.91 nH |

### 4.4 Simulation

Circuit simulations were performed in **LTspice**, validating the design prior to hardware prototyping. GaN MOSFET models were imported to accurately represent real device behavior, including parasitic elements and non-ideal characteristics. Square-wave voltage sources mimicking gate driver outputs (fast rise/fall times, proper dead-time) were used to drive the switches.

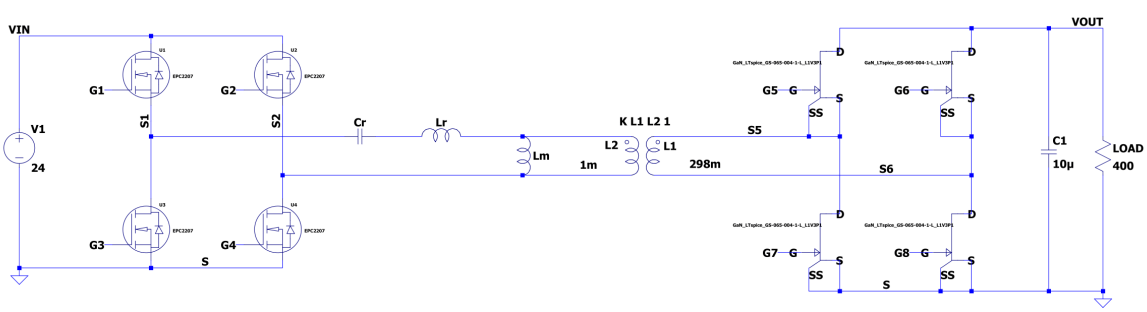



Figure 11: Complete LLC converter schematic in LTspice.

The schematic (Figure 11) is organized as follows:

- **Left side:** Low-voltage, high-current primary-side GaN MOSFETs in full-bridge configuration, driven by the PV panel voltage.
- **Center:** LLC resonant tank with $C_r$, $L_r$, and $L_m$ values taken from Table 3, along with the step-up transformer sized for the required boost ratio.

- **Right side:** High-voltage, low-current synchronous full-bridge rectifier using GaN MOSFETs instead of diodes, reducing forward-drop conduction losses thanks to low $R_{DS(on)}$.
- **Output:** Bulk filter capacitor and resistive load.

The PWM gate signals are configured as follows:

Table 4: PWM generator parameters.

| **Parameter** | **Generator A** | **Generator B** |
|---|---|---|
| $V_{initial}$ | 0 V | 0 V |
| $V_{on}$ | 5 V | 5 V |
| $T_{delay}$ | 0 s | $\frac{1}{2f} + 2t_d$ |
| $T_{rise}$ | 4 ns | 4 ns |
| $T_{fall}$ | 4 ns | 4 ns |
| $T_{on}$ | $\frac{1}{2f}$ | $\frac{1}{2f} - 4t_d$ |
| $T_{period}$ | $\frac{1}{f}$ | $\frac{1}{f}$ |

The 5 V drive amplitude is consistent with GaN MOSFET $V_{GS}$ requirements. Rise and fall times of 4 ns reflect typical gate driver specifications. Parameters $f$ (switching frequency) and $t_d$ (dead time) are swept for each simulation run.

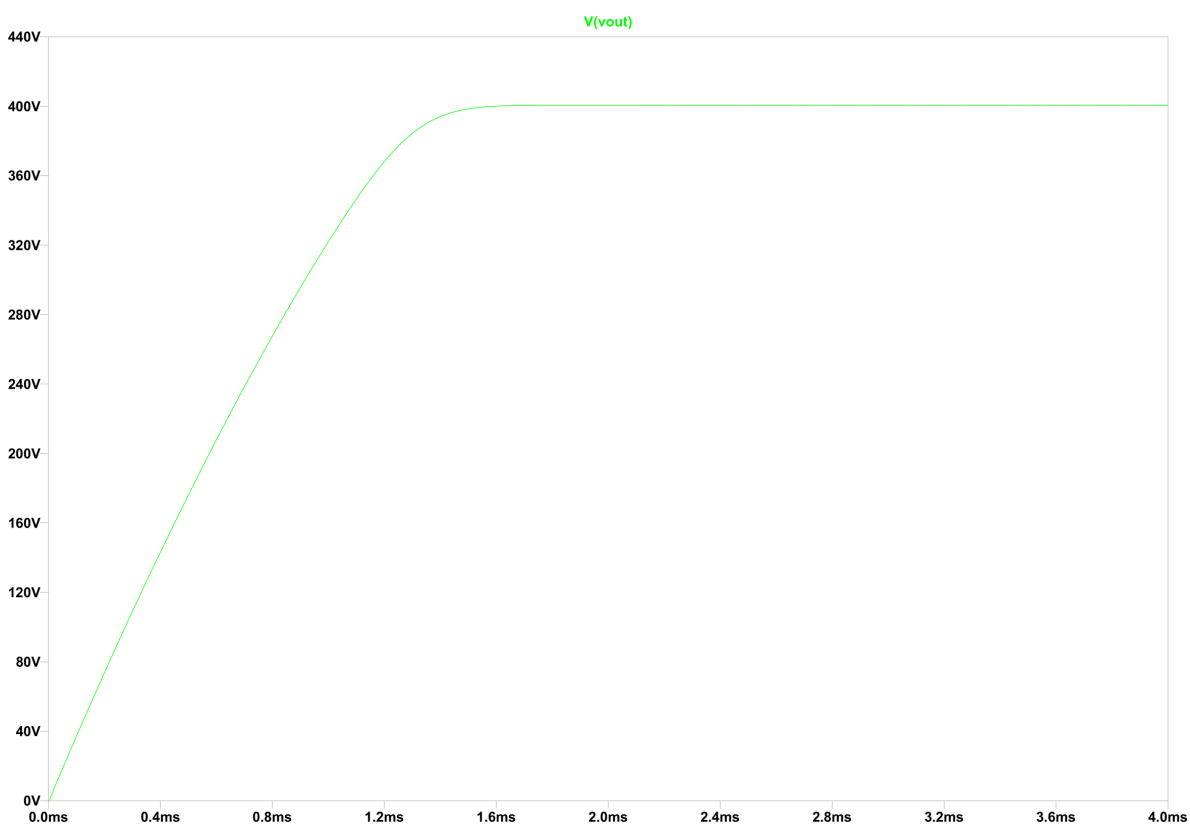



Figure 12: Simulated output voltage waveform.

Efficiency is computed using LTspice's built-in measurement directives:

- `.meas Pin AVG V(VIN)*I(V1)`
- `.meas Pout AVG V(VOUT)*I(LOAD)`
- `.meas Eff param Pout/Pin`

Table 5: LLC converter simulated efficiency.

| $f_{sw}$ | $P_{in}$ (W) | $P_{out}$ (W) | Efficiency (%) |
|---|---|---|---|
| 100 kHz | 408.91 | 400.25 | 97.88 |
| 500 kHz | 419.34 | 400.57 | 95.52 |
| 1 MHz | 427.03 | 400.31 | 93.74 |

A clear efficiency decline is observed with increasing switching frequency. The primary loss mechanisms are:

- **Passive component losses:** At higher frequencies, capacitor dielectric losses and inductor core/winding losses increase substantially.
- **Active device losses:** Even under soft switching, switching transitions represent a larger fraction of the total period at high frequencies, increasing average switching losses.
- **Transformer core losses:** Hysteresis and eddy-current losses in the magnetic core scale with frequency, becoming significant above 500 kHz.

However, higher switching frequency yields directly proportional reductions in passive component values, enabling smaller, lighter, and less expensive magnetics and capacitors — an important system-level trade-off.

## 5 Complete Micro-Inverter

### 5.1 Full System Simulation

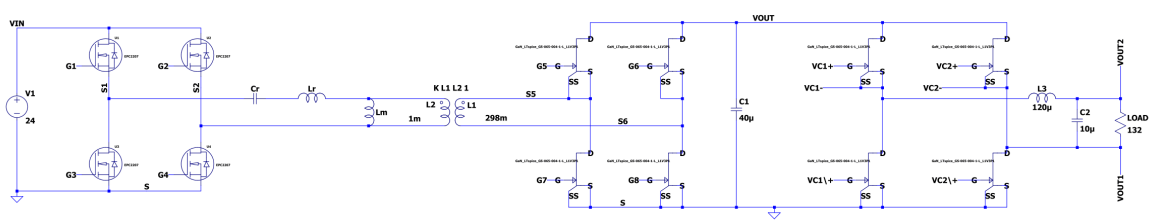



Figure 13: Complete micro-inverter schematic.

The complete micro-inverter (Figure 13) appends a full-bridge H-bridge inverter stage to the LLC DC-DC converter output. The H-bridge converts the 400 V DC bus to AC, followed by an LC output filter.

The LC filter cutoff frequency is:

$$f_c = \frac{1}{2\pi\sqrt{LC}}$$

The filter is designed so that $f_c$ satisfies $f_{out} = 50\,\text{Hz} \ll f_c \ll f_{sw}$, effectively attenuating switching harmonics while passing the fundamental. Higher switching frequencies allow smaller $L$ and $C$ values for the same cutoff frequency.

**Unipolar PWM modulation** is implemented using op-amp comparators that compare a sinusoidal reference with a high-frequency triangular carrier. Two complementary PWM signals (with dead time) drive each inverter leg:

- Leg A: comparator output of $\sin(\omega t)$ vs. triangular carrier.
- Leg B: comparator output of $\sin(\omega t + 180°)$ vs. triangular carrier.

The frequency modulation ratio is:

$$m_f = \frac{f_s}{f_1}$$

A higher $m_f$ improves output waveform quality. The amplitude modulation ratio is:

$$m_a = \frac{v_{control}}{v_{triangular}}$$

Values of $m_a$ near or exceeding 1.0 drive the inverter into the over-modulation (non-linear) region, degrading output quality and efficiency.

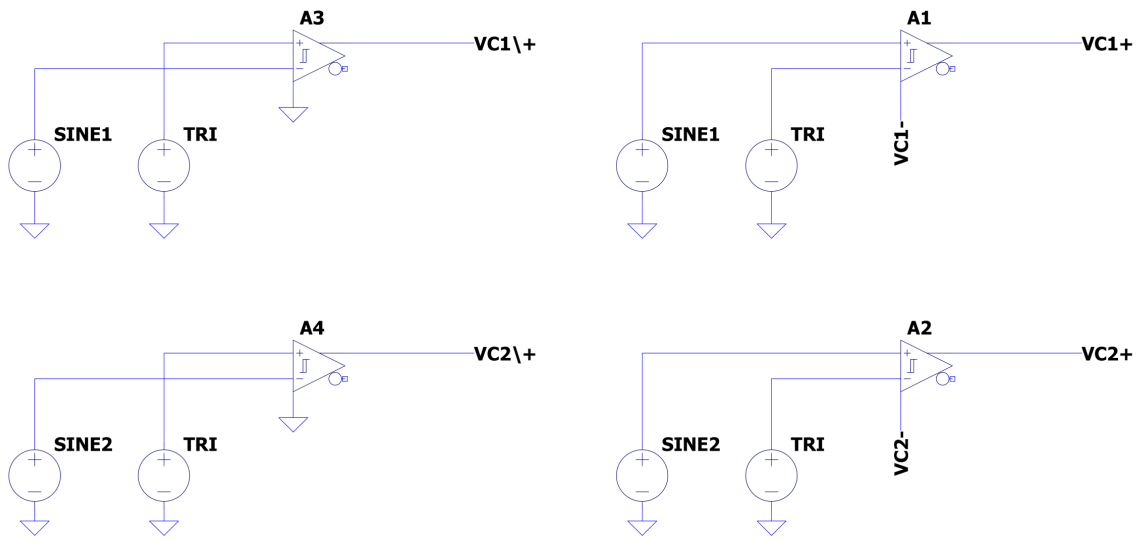



Figure 14: PWM generation comparators (unipolar modulation).

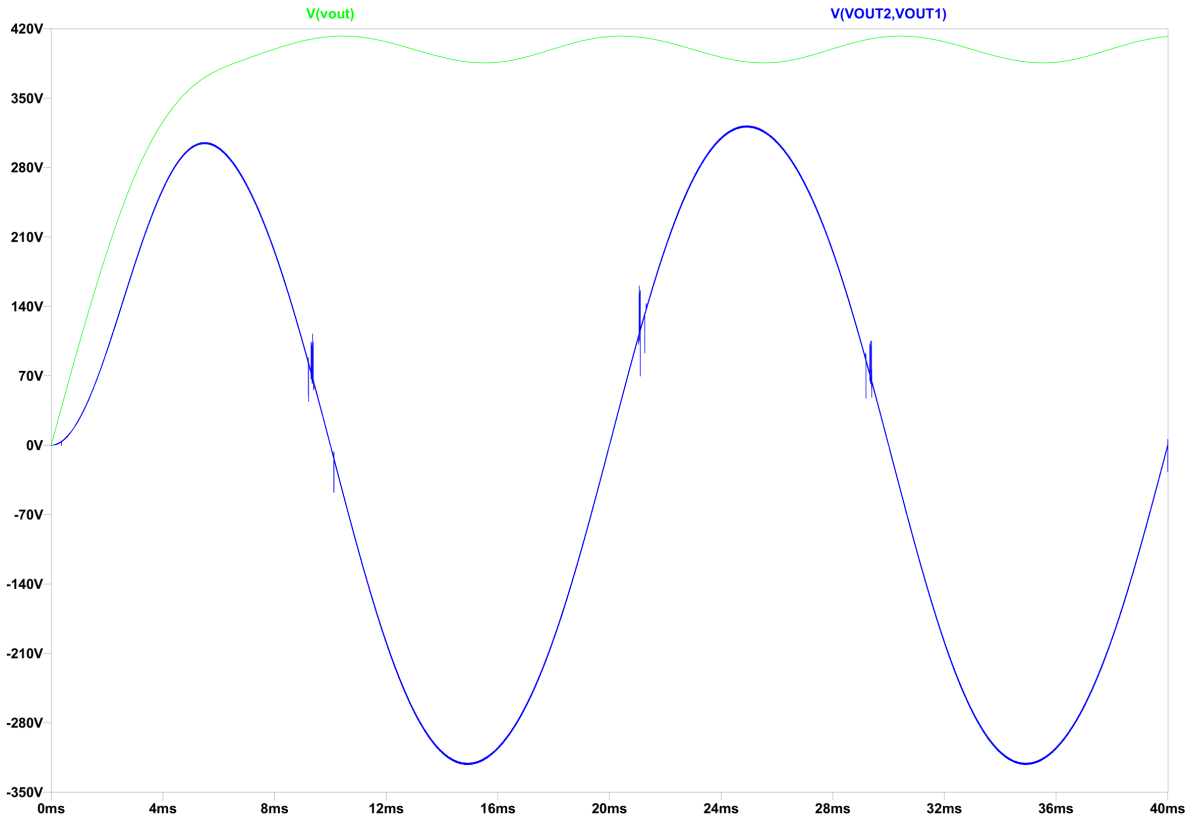



Figure 15: Micro-inverter output sinusoidal waveform (50 Hz, 230 $V_{RMS}$).

A brief transient distortion is visible at startup as the LLC resonant tank charges and the system stabilizes. Steady-state operation produces a clean 50 Hz, 230 $V_{RMS}$ sinusoidal output at 400 W.

Table 6: Micro-inverter system efficiency breakdown.

| **Switching Frequency** | **LLC Eff. (%)** | **Inverter Eff. (%)** | **System Eff. (%)** |
|---|---|---|---|
| 100 kHz | 97.88 | 98.79 | 94.71 |
| 500 kHz | 95.52 | 97.65 | 92.64 |
| 1 MHz | 93.74 | 95.76 | 88.70 |

Figure 16 illustrates efficiency as a function of output power for each switching frequency. Peak efficiency is observed in the 150 W to 250 W range at 100 kHz, 250 W to 350 W at 500 kHz, and 300 W to 400 W at 1 MHz. These peaks reflect the interaction between conduction losses (dominant at heavy load) and switching losses (dominant at light load), and underscore the importance of matching the switching frequency to the expected load profile.

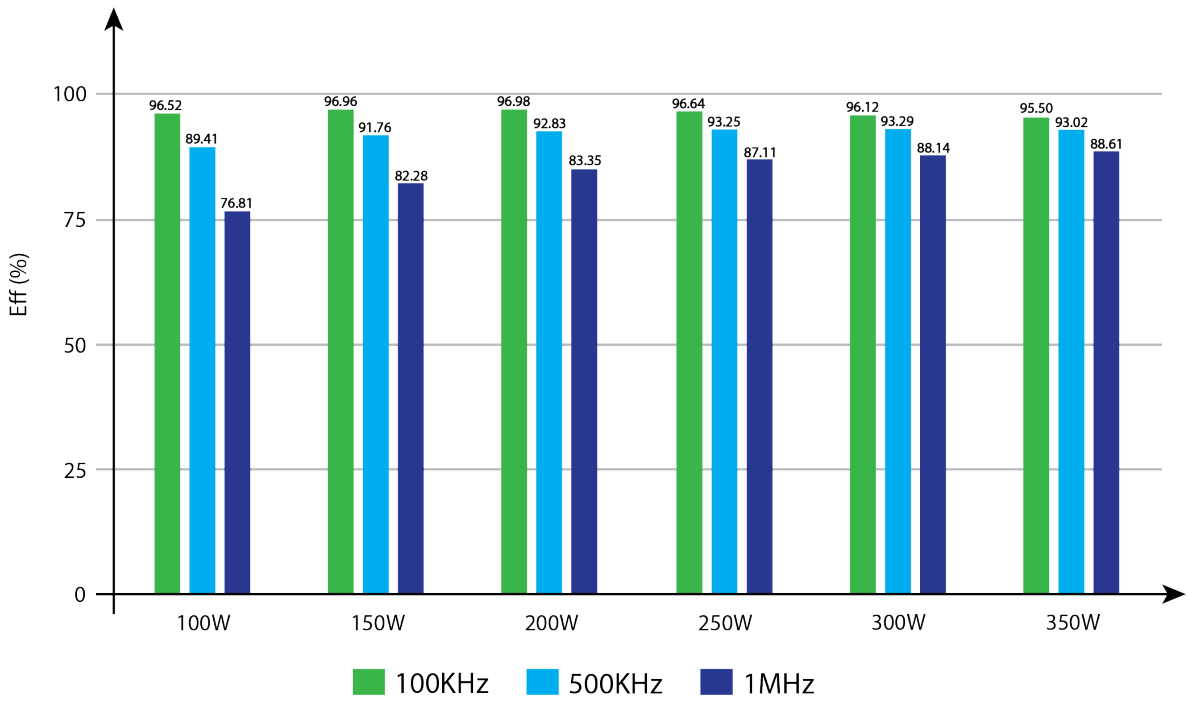



Figure 16: System efficiency vs. output power for three switching frequencies.

## 6 Conclusion

This report presented the detailed design and simulation of a GaN-based LLC step-up converter for a photovoltaic micro-inverter application. The study covered the full design flow: from topology selection and component sizing to LTspice validation and system-level efficiency analysis.

The LLC resonant converter was confirmed as the highest-efficiency topology among the evaluated alternatives, achieving up to **97.88%** efficiency at 100 kHz. The adoption of GaN MOSFETs enabled high-frequency operation with low switching and conduction losses, a combination that would be difficult to achieve with conventional Si devices at this power level.

System-level simulation demonstrated that the complete micro-inverter achieves a **94.71%** end-to-end efficiency at 100 kHz while delivering a clean 50 Hz / 230 $V_{RMS}$ sinusoidal output at 400 W. The efficiency–frequency trade-off was quantified: higher switching frequencies reduce passive component size and cost but increase losses, with system efficiency dropping to **88.70%** at 1 MHz.

Future work could investigate adaptive frequency control strategies for peak-efficiency tracking across the full load range, advanced magnetic design optimized for high-frequency GaN operation, and thermal modeling to validate reliable operation across environmental conditions. The methodologies and results presented here provide a solid reference for future GaN-based micro-inverter development targeting the residential and commercial renewable energy markets.

## 6 References

[1] Meshael, H., Elkhateb, A., and Best, R. Topologies and design characteristics of isolated high step-up dc–dc converters for photovoltaic systems. *Electronics* (2023), 42.

[2] Technologies, I. Application note: Resonant llc converter operation and design. Tech. rep., Infineon Technologies, 2012.